Greenhouse Anomaly Detection in Environmental Sensor Data Using TinyML

Zhengliang Song

October 2025

Contents

1 Introduction **1**

1.1 Background 1

1.2 Aim and objectives 2

1.2.1 Aim 2

1.2.2 Objectives 2

1.3 Product Overview 2

1.3.1 Scope 2

1.3.2 Audience 3

2 Background Review **4**

2.1 Existing Approaches 4

2.2 Related Literature 4

3 Methodology **6**

3.1 Approach 6

3.2 Technology 7

3.3 Version Management 7

4 Project Management **8**

4.1 Activities 8

4.2 Schedule 9

4.3 Data Management 10

4.4 Deliverables 10

A Ethics Form **14**

Chapter 1

Introduction

1.1 Background

In modern agriculture, intelligent farming has become an important part, which deploys Internet of Things (IoT) technology, artificial intelligence (AI), and edge computing to maximize crop yields, reduce wastage of resources, and monitor environment parameters [3], [4]. Environmental sensors also form a part of the process, since they provide important data relating to temperature, humidity, etc. in smart greenhouse.

These sensors suffer from faults, outliers, and calibration drift and may have inaccurate or misleading data. Real-time anomaly detection deployment underpins data assurance and preserving decision-making accuracy. Conventional methods of anomaly detection also depend on central servers or on-cloud computation, which may be inefficient or inapplicable in remote or energy-constraint applications.

TinyML, which involves the deployment of machine learning algorithms on microcontrollers with incredibly minimal power consumption, offers a potential solution to supporting anomaly detection at the edge. This research examines the development of a TinyML-based anomaly detection system suitable for environmental sensor data, based on the Arduino Nano 33 BLE Sense board, aiming to realize autonomous, real-time detection of faults in intelligent agriculture systems [1], [6].

1.2 Aim and objectives

1.2.1 Aim

Design, create, and test a light-weight machine learning model using TinyML to identify anomalies in environment sensor data related to intelligent agriculture on a microcontroller.

1.2.2 Objectives

- Conduct a literature review on TinyML in anomaly detection and intelligent farming.
- Retrieve and preprocess environment sensor data using Arduino and Python.
- Develop and train an anomaly detection model with microcontroller-compatible lightweight ML algorithms.
- Deploy and optimize the model using TensorFlow Lite for Microcontrollers on the Arduino Nano 33 BLE.
- Design and perform experiments to conform to accuracy, latency, and power efficiency.
- Assess the system's practicality in real life, scalability, and shortcomings.

1.3 Product Overview

1.3.1 Scope

These include a proof-of-concept board that gathers environment parameters like temperature, humidity, etc. It locally runs an anomaly detection script on the Arduino Nano 33 BLE Sense board with TinyML technology. This identifies anomalies, such as real-time sensor spikes, hardware failure, or abnormal trends.

1.3.2 Audience

This project is artifacted for multiple audiences with interests in embedded systems, Edge AI, and smart agriculture. The following stakeholders may benefit directly from the system and its results:

1. Agricultural technology researchers and practitioners - who are seeking cost-effective, autonomous solutions for monitoring and responding to environmental anomalies in greenhouses.

2. Embedded systems engineers and TinyML developers - who are exploring real-world applications of machine learning on resource-constrained devices.

3. Academic supervisors and examiners - who will evaluate the project based on innovation, feasibility, implementation, and relevance to real-world problems.

4. IoT educators and students - who can use the project as a case study to learn about TinyML workflows, model deployment, and sensor integration in edge devices.

In addition, the results of this project may also apply on future implementations of offline anomaly detection in environmental monitoring systems, especially in scenarios where network access is limited or power efficiency is critical.

Chapter 2

Background Review

2.1 Existing Approaches

Traditional anomaly detection methods of sensor systems rely on statistical thresholding, smoothing of the signal, or far-end ML algorithms deployed in the cloud. While good, they suffer from the limitation of latency, always-on connectivity, and high-bandwidth consumption. Edge computing with TinyML provides a alternative by moving analytics closer to the data source [5], [9].

For example, Kulkarni and Bhudhwale applied TinyML for real-time environmental monitoring [3], while Ihoume et al. developed a multi-label TinyML model for greenhouse microclimate control [2]. Hidalgo and Vidal focused on TinyML for irrigation [4]. Most solutions focus on control and prediction, not on sensor reliability. Faulty readings can lead to poor decisions even in well-optimized systems. Hence, anomaly detection becomes an enabling layer for more resilient smart farming.

2.2 Related Literature

Antonini et al. proposed an unsupervised TinyML system for hostile environments [1], and this project model was inspired by this. Ahmed et al. integrated unsupervised models into intelligent farm-based IoT systems [10].

These researchers employed TinyML across LoRa networks to perform anomaly

detection at the edge [5], and Tsoukas et al. investigated how to construct TinyML on constrained hardware [6]. Anwar et al. investigated drone-enabled intelligent farming with TinyML and transfer learning [7]. Vieira investigated in-device evapotranspiration prediction, which applies to seasonal anomaly detection [8]. The methodology they employ regarding unsupervised autoencoders holds significant importance for practical implementation in scenarios where annotated data is limited. This consideration guides the selection of the model architecture pertinent to this study.

Chapter 3

Methodology

3.1 Approach

It will adopt a hybrid development model:

- Research Stage: Scoping of technology and literature, and data exploration.
- Prototyping Phase: Access environment data and create preliminary models using Python.
- Development Phase: Transform models to TinyML and run them on the Arduino board.
- Evaluation Phase: Assess performance (accuracy, memory, latency, energy).

The project follows an unsupervised mainline suited to scarce labels: compact autoencoders with k-means or simple statistical thresholds on reconstruction-error scores. A tiny supervised MLP may be trained only as a comparative baseline if a small labelled slice exists; it is not part of the deployed solution.

Architecture is modular to allow fast iteration and on-device testing. We perform per-sensor normalization and sliding windows. We quantize and memory-profile our selected model such that peak SRAM less than 200 KB and single-inference latency less than 200 ms; Flash usage stays in device limits.

3.2 Technology

- Arduino Nano 33 BLE Sense (onboard sensors: temperature, humidity, light).
- Python (Pandas, NumPy, Scikit-learn, TensorFlow) to set up the data and train the model.
- TensorFlow Lite for Microcontrollers (TFLM) to convert and run models.
- Arduino IDE + Edge Impulse CLI to deploy and debug the firmware.
- Dataset: Initially synthetic data from the sensors and real sensor logs collected over a number of days.
- Modeling: Try small autoencoders, statistic-based thresholding, and 1D CNNs if feasible.

3.3 Version Management

Version control with GitHub, in:

- data – both processed and raw data records.
- models – converted and trained models.
- code – Arduino and Python code.
- docs – reports and diagram.
- Commitments will be organized per milestone (data gathering, model training, deployment, assessment).

Chapter 4

Project Management

4.1 Activities

The following activities are designed to meet the objectives stated in Section 1.2.2, with a clear path to achieve the overarching project aim: developing an embedded anomaly detection system using TinyML.

1. **Literature review** – Review current methods in TinyML, anomaly detection, and smart agriculture.
2. **Hardware setup** – Configure Arduino Nano 33 BLE Sense with sensors and test I/O.
3. **Data collection** – Use Edge Impulse to record temperature, humidity, and light readings in varied environments.
4. **Feature engineering** – Apply sliding window techniques and extract statistical features.
5. **Model selection and training** – Train compact autoencoders; derive scores from reconstruction error and combine with k-means/statistical thresholds. Train a tiny MLP baseline only if labels exist; not part of the deployed solution.
6. **Quantization and deployment** – Convert models to int8 and deploy to microcontroller.

7. **Testing and validation** – Test both model performance and real-time device behavior.
8. **Documentation and reporting** – Prepare final report, codebase documentation, and reflective evaluation.
9. **Optional BLE extension** – If time permits, integrate BLE-based notification system.

4.2 Schedule

Activities above will be executed over two semesters, using an iterative Agile-like workflow. The high-level timeline is illustrated in the Gantt chart (Appendix or separate file), summarised as:

- **Semester 1 (Weeks 1–12)** – Research, hardware setup, data collection, initial model training, and basic deployment.
- **Semester 2 (Weeks 13–24)** – Evaluation, testing, documentation, optional BLE expansion, and final report writing.

Milestones are tracked weekly, using GitHub issues and changelog tags. Internal reviews are done bi-weekly for alignment with objectives.

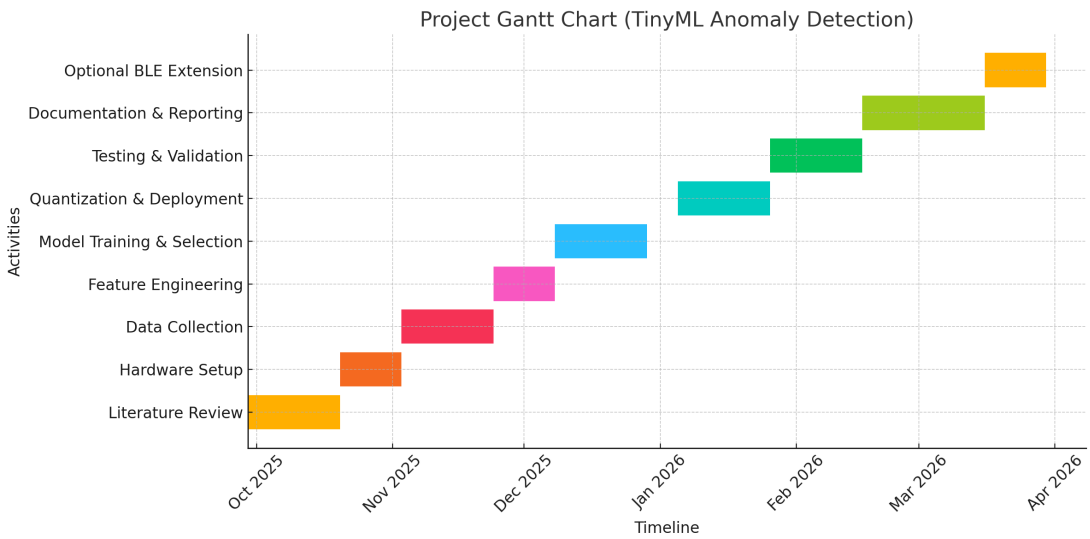

Figure 4.1: Gantt Chart Overview

4.3 Data Management

Project and artifact data will be tracked as follows:

Project Data (Logs, Code, Reports)

- **Codebase**: Managed via GitHub private repository with commit history and branches.
- **Progress logs**: Maintained in a markdown journal and GitHub issues.
- **Reports**: Proposal drafts, interim reviews, and final report.

Artifact Data (Sensor Readings, Models)

- **Sensor datasets**: Captured via Edge Impulse Data Forwarder and backed up in versioned folders.
- **ML Models**: Saved at each stage (baseline, tuned, quantized) with appropriate versioning.

Legal and ethical considerations are minimal since no personal data is collected; all data is environmental and stored locally.

4.4 Deliverables

The following deliverables are expected during the course of the project:

Deliverable	**Description**
Proposal Report	Completed planning document covering aims, methodology, feasibility, etc.
Sensor Dataset (Raw & Processed)	Collected environmental readings, labeled for training/testing
Trained TinyML Models	Autoencoder and MLP models (quantized TFLite version)
Embedded Codebase	Arduino-compatible source code for inference, alerting, and testing
System Demo (Video)	Demonstration of working system with real-time alerts
Final Report	Complete academic report including evaluation and reflection
Project Repository	GitHub repo with logs, code, documentation, and version control
Optional BLE Alert Feature	Bluetooth interface for mobile anomaly notifications (if time permits)

Bibliography

[1] M. Antonini, M. Pincheira, M. Vecchio, and F. Antonelli, "An adaptable and unsupervised TinyML anomaly detection system for extreme industrial environments," *Sensors*, vol. 23, no. 4, p. 2344, 2023.

[2] I. Ihoume, R. Tadili, N. Arbaoui, and M. Benchrifa, "Developing a multi-label TinyML model for optimized greenhouse microclimate control from multivariate sensed data," *Artificial Intelligence in Agriculture*, vol. 6, pp. 87–100, 2022.

[3] S. Kulkarni and S. Bhudhwale, "Real-time environmental monitoring in smart agriculture using TinyML and machine learning," in *Proc. IEEE Int. Conf.*, 2024.

[4] C. H. Hidalgo and A. González-Vidal, "Edge computing in smart agriculture based on TinyML for irrigation control," in *Proc. IEEE 9th World Forum on Internet of Things*, 2023.

[5] A. Karras, C. Karras, and A. Giannaros, "TinyML-based event detection: An edge-cloud approach for smart agriculture over LoRa WSNs," in *Proc. IEEE SEEDA-CECNSM*, 2023.

[6] V. Tsoukas, A. Gkogkidis, and A. Kakarountas, "Elements of TinyML on constrained resource hardware," in *Computing and Data Science*. Springer, 2022, pp. 305–320.

[7] A. M. Hayajneh, S. A. Aldalahmeh, and F. Alasali, "TinyML on the edge: Transfer learning for UAV-assisted smart farming," *IET Smart Cities*, 2024.

[8] M. J. O. Vieira, "On-device machine learning for forecasting reference evapotranspiration," Ph.D. dissertation, Universidade Federal de Alagoas, 2024. [Online]. Available: `https://www.repositorio.ufal.br/handle/123456789/16887`

[9] Z. E. Ahmed, A. A. Hashim, and R. A. Saeed, "TinyML network applications for smart cities," in *TinyML for Edge Intelligence*. Elsevier, 2024.

[10] M. F. Almufareh, M. Humayun, Z. Ahmad, and A. Khan, "IoT monitoring and control in smart farming with anomaly detection using unsupervised ML," *IEEE Access*, 2024.

Appendix A

Ethics Form

OXFORD BROOKES UNIVERSITY

Faculty of Technology, Design and Environment - Ethics Review Form E1

- This form should be completed jointly by the **Supervisor and Student** who is undertaking a research/major project which involves human participants.
- It is the **Supervisor** who is responsible for exercising appropriate professional judgement in this review.
- Before completing this form, please refer to the University **Code of Practice for the Ethical Standards for Research involving Human Participants**, available at https://www.brookes.ac.uk/sites/research-support/research-ethics-and-integrity/policies-procedures-and-useful-documents/obu-urec-code-of-practise-policies and to any guidelines provided by relevant academic or professional associations.
- Note that the ethics review process needs to fully completed and signed **before fieldwork commences**.

(i) Project Title:

GreenHouse Anomaly Detection in Environmental Sensor Data Using TinyML

(ii) **Name of Supervisor and School in which located:**

Dr. Abusaleh Jabir, School of Engineering, Computing and Mathematics

(iii) Name of Student and Student Number:

Zhengliang Song 19382584

(iv) Brief description of project outlining where human participants will be involved (30-50 words):

None

		Yes	No
1.	Does the study involve participants who are unable to give informed consent (e.g. children, people with learning disabilities)?	☐	☑
2.	If the study will involve participants who are unable to give informed consent (e.g. children under the age of 18, people with learning disabilities), will you be unable to obtain permission from their parents or guardians (as appropriate)?	☐	☑
3.	Will the study require the cooperation of a gatekeeper for initial access to groups or individuals to be recruited (e.g. students, members of a self-help group, employees of a company)?	☐	☑
4.	Are there any problems with the participants' right to remain anonymous, or to have the information they give not identifiable as theirs?	☐	☑
5.	Will it be necessary for the participants to take part in the study without their knowledge/consent at the time? (e.g. covert observation of people in non-	☐	☑

	public places?)		
6.	Will the study involve discussion of or responses to questions the participants might find sensitive? (e.g. own traumatic experiences)	☐	☑
7.	Are drugs, placebos or other substances (e.g. food substances, vitamins) to be administered to the study participants?	☐	☑
8.	Will blood or tissue samples be obtained from participants?	☐	☑
9.	Is pain or more than mild discomfort likely to result from the study?	☐	☑
10.	Could the study induce psychological stress or anxiety?	☐	☑
11.	Will the study involve prolonged or repetitive testing of participants?	☐	☑
12.	Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?	☐	☑
13.	Will deception of participants be necessary during the study?	☐	☑
14.	Will the study involve NHS patients, staff, carers or premises?	☐	☑

Signed:		**Supervisor**
Signed:		**Student**
Date:	Oct.16 2025	

What to do now:

1. If you have answered **'no'** to all the above questions:

   (a) The student must **send** the completed and fully signed E1 form to their **Dissertation Module Leader.**
   (b) The student must keep a copy of the E1 form which must be bound into their dissertation as an appendix.
   (c) The supervisor must keep a copy of the E1 form as they are responsible for monitoring compliance during the fieldwork.

2. If you have answered **'yes'** to **any** of the above questions:

   (a) The supervisor and student must complete the TDE E2 form available at https://www.brookes.ac.uk/sites/research-support/research-ethics-and-integrity/research-ethics/ethics-application-process-forms/forms-and-templates-for-masters-and-undergraduate
   (b) Note that the information in the E2 must be in **sufficient detail** for the ethical implications to be clearly identified.
   (c) The signed E2 and signed E1 Form must be emailed to Tim Jones (tjones@brookes.ac.uk) who is the Faculty Research Ethics Officer (FREO) for review. Please allow **at least two weeks** for this review process.
   (d) If/when approved the FREO will issue an E3 Ethics Approval Notice.
   (e) The student must send the E1, E2 and E3 Notice **to the Dissertation Module Leader**.
   (f) The student must also keep copies which must be bound into their dissertation as an appendix.
   (g) The supervisor must keep a copy of documentation to monitor compliance during field work.

3. If you answered 'yes' to any of questions 1-13 and 'yes' to question 14, an application must be submitted to the appropriate NHS research ethics committee. This is an onerous and time consuming process so the supervisor should liaise early with the FREO if the student is considering this.